

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant and Teaching Assistant Aug 2025 to May 2026

- Supported **200+** students as a TA in Programming Paradigms, AI, and Machine Learning through debugging sessions, code review, and grading.
- Reduced rebuffering by **19%** versus a fixed-rate baseline by building a **Docker**, **WebTransport**, and **Media-over-QUIC** testbed and running controlled adaptation experiments across network and protocol configurations.
- Cut experiment setup effort by about **55%** by automating orchestration, configuration, **telemetry** collection, and analysis, making large experiment sweeps reproducible.
- Built and evaluated **RAG** pipelines (chunking, embeddings, vector retrieval, top-k, grounded prompts), separating **retrieval** failures from generation failures to guide fixes.

Turito | Software Engineering Intern, EdTech Platform Jun 2022 to Oct 2022

- Improved reliability of a live **online classroom** platform (WebRTC, RTP/RTCP, about 12 regions) where dropped sessions directly interrupted lessons for students and teachers.
- Cut controlled-disruption failures by **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Slashed p95 **CPU** by **22%** by profiling high-frequency media paths and removing excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Reduced production diagnosis time by **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Reduced performance-investigation time by about **40%** with tooling built on Rust workers, **Python** aggregation, **SQLite**, Flask, and Chart.js.

TECHNICAL SKILLS

Languages: Python, Java, C++, Rust, TypeScript, SQL
Backend & Distributed: FastAPI, Flask, REST, gRPC, Kafka, Temporal, WebSockets, asyncio, Tokio
Data & Cloud: PostgreSQL, Redis, SQLite, MongoDB, Docker, AWS, Linux, CI/CD, pytest
Frontend & AI: React, TypeScript, Next.js, PyTorch, RAG, OpenAI and Anthropic APIs

PROJECTS

AI Learning Hub, Grounded RAG | *Python, FastAPI, LangChain, OpenAI, Docker, AWS*

- Built grounded **RAG** over **10,000+** articles with source citations and abstention when retrieved evidence was insufficient.

AutoGrade AI, Multi-Agent Evaluation Platform | *Python, Temporal, Kafka, gRPC, PostgreSQL, Qdrant*

- Built rubric-driven grading with multiple LLM agents, repeated evaluation to expose disagreement, confidence gates, human-review queues, and audit traces over **Temporal** workflows.

ReelRank, Two-Stage Recommender | *PyTorch, C++ HNSW, FastAPI, React, TypeScript*

- Built a two-tower retrieval model with ANN candidate generation and a neural ranker, served through **FastAPI** and a **React/TypeScript** UI with metadata-aware cold start.

OPEN SOURCE

- microsoft/markitdown**: fixed EPUB conversion for percent-encoded manifest paths, with regression tests (merged)
- fission/fission**: 3 merged PRs, Kubernetes event and KEDA object capture in support dumps, removed duplicate package enumeration

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Distributed Computing, Advanced Algorithms, Database Systems, Computer Networks, Machine Learning, NoSQL

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)